Project Instructions: Comparative Analysis of GO-Enriched Communities in Bacterial STRING Networks

Definition: Module.

*In this project, **a module** is a densely connected group of proteins within a STRING interaction network that is inferred to represent a shared biological function or process.* **(Additional explanations are given Questions&Answers document).**

Project Overview

This mini project combines network analysis, module detection, functional enrichment, and biological interpretation to identify and compare functional modules across and within bacterial species. To this end, you are expected to analyze bacterial protein-protein interaction networks derived from STRING and compare functional modules across species. Your goals are (1) to identify intra-species (within-species) modules, (2) annotate them with functional information, and (3) detect functionally conserved modules, both across different bacteria and within the same bacteria.

What students are expected to do

Part 1: In the first part of the project, you should work with bacterial STRING interaction networks together with functional annotation files for the proteins in each network. For each bacterium, construct or load the interaction network, inspect the weighted edges, and identify modules or communities inside the network: **References to various optional algorithms for this task, and to their implementations in python, are given in the Questions&Answers document.** The project slides show k-means as one example for a network partition, but you are to select any method that, in your opinion, makes sense for protein-protein network graph structure and module discovery. Consider\try several approaches for module discovery, then explain in your report which approach you chose and why, and report the parameter settings\cutoffs you chose.. **(Some of the approaches are reviewed in the Questions & Answers document available in Moodle).**

After detecting modules within each bacterial network in the input files, you should **characterize and annotate each module** using GO enrichment or equivalent functional annotation summaries **(details and references for the statistical analysis are given in the Questions&Answers document).** The purpose of this step is to assign biological meaning to each detected sub-network. A densely connected set of proteins may represent e.g., an efflux pump, membrane-associated process, transport system, or another functional unit. Explain, in your report, which functional category descriptors you considered, which statistical tests you employed for functional enrichment tests and what were the parameters\threshold-cutoffs you chose.

Part 2. In the second part of the project, the objective is to **compare** ("all against all") modules, both across bacteria and within bacteria, **using their enriched functional descriptors rather than exact protein identity alone.** Represent each module by its enriched biological terms. Then, compare modules across species and within species, using a set-similarity metric such as the Jaccard index exemplified in the project slides. This comparison should help you identify functionally conserved modules across species, functionally conserved modules within species, and singular specie-specific functional modules.

For the comparison, run at least two separate experiments: B5 versus B6 (Input 1) and B21 versus B22 (input 2). You are encouraged to analyze additional networks as part of your unique contribution.

Datasets currently available in Moodle: the two bacteria from file Input1 and the two bacteria from dataset Input2 (found as zip files in Moodle) are described below:

B5: Salmonella Virulence Network.

B6: Vibrio Cholera Virulence Network.

B21: Bacillus Subtilis Resistance Network.

B22: Erwinia Resistance Network.

Explain, in your report, how you computed the similarity between modules and what were the parameters\ threshold-cutoffs you chose. Did you need to use different parameters\threshold cutoffs in order to discover cross-species conserved modules for the Input1 set, in comparison to the Input2 set? Analyze your results in the context of the characteristics and habitats of the input bacteria.

In this part of the project, each group (consisting of either a single student or two students at most), is expected to propose and implement a "creative" approach to the solution of the project objectives. Explain, in your report, what you did to make your solution unique, and how it affected the resulting discoveries.

What the final submission should contain

Part 1: Your final submission should include a Google Collab notebook with the code\script to implement the required objectives for this part.

In addition to the code, your report should clearly describe the organisms and datasets analyzed, the network and annotation inputs, the method used for module detection, and the method used for functional enrichment (including the values you used to set parameter\threshold cutoffs). It should present (using visual graphics) the detected modules (presented as networks) for each bacterium together with their major enriched GO terms or other functional descriptors.

In your report, include all the discussion points regarding Part 1 that are mentioned and highlighted in blue in the previous section.

Code (documented Jupiter notebook) and report due date: 19\04\2026

Part 2:

In addition to the code, your submission should also include an all-against-all module comparison section, separately for the bacteria in Input1 and the bacteria in Input 2, showing which modules are functionally conserved across the input bacteria, and which modules are functionally conserved within input bacterias. In your report, you should also explain how similarity was measured in your implementation, and what biological conclusions can be drawn from both conserved and unconserved modules. Include figures or tables to illustrate your results. At minimum, your report or presentation should contain network and module visualizations, enriched-term summaries, a table of conserved module matches across and within species.

In your report, include all the discussion points regarding Part 2 that are mentioned and highlighted in blue in the previous section.

Code (documented Jupiter notebook) and report, due date:21\06\2026

**** Note that your programs should read all the input files from the current directory (from which the program is run via Google Collab) and save the output files also to the current directory.**

**** You may use AI tools in any way that benefits the resulting programs, but the programs you deliver should fully implement Objective 1 and Objective 2 – and should produce the desired output based on the given input in a reproducible way.**